

MA 102 MATHEMATICS II
Ordinary Differential Equations
Lecture 2
(March 5, 2024)



An equation involving one dependent variable and its derivatives with respect to one or more independent variables is called a **differential equation**.

When a differential equation contains only ordinary derivatives, it is called an **ordinary differential equation** while a differential equation is called a **partial differential equation** if it contains partial derivatives.

This means that an ODE is a differential equation which may contain the dependent variable (say, y), the independent variable (say, x) and the derivative(s) of the dependent variable. While an ODE **must contain derivative(s)**, it is not necessary for it to contain x and/or y explicitly.

Order and degree of a differential equation

The **order** of a differential equation is the order of the highest derivative present.

The **degree** of a differential equation is the degree (power) of the highest derivative present.

It may be noted that the degree of a differential equation may not always be defined whereas the order of it is always defined as a certain number. The degree of any given differential equation can be obtained only when its highest derivative is in the form of a polynomial. In other words, the degree cannot be determined if the highest derivative is expressed not as a polynomial.

Example 1. $\left(\frac{dy}{dx}\right)^2 + \sin\left(\frac{d^3y}{dx^3}\right) = 0$, **Example 2.** $\frac{dy}{dx} + \exp\left(\frac{d^2y}{dx^2}\right) = 0$

However, for the differential equations

Example 3. $\frac{d^4y}{dx^4} + \sin\left(\frac{d^3y}{dx^3}\right) = 0$, **Example 4.** $\left(\frac{d^3y}{dx^3}\right)^2 + \exp\left(\frac{d^2y}{dx^2}\right) = 0$,
the degrees are, respectively, 1 and 2.

The most general ODE of n -th order has the form

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0, \quad (1)$$

where F is some arbitrary function at this moment.

In simple notations for derivatives, (1) can be written as

$$F(x, y, y', y'', \dots, y^{(n)}) = 0. \quad (2)$$

A general linear n -th order ODE can be written as

$$a_0(x)y^{(n)} + a_1(x)y^{(n-1)} + \dots + a_{n-1}(x)y' + a_n(x)y = 0, \quad (3)$$

so that no term contains a product of y and any of its derivatives.



The most general first-order ODE has the form

$$F(x, y, y') = 0. \quad (4)$$

Generally speaking, it is very difficult to solve first-order differential equations. Even a simple-looking equation

$$\frac{dy}{dx} = f(x, y) \quad (5)$$

cannot be solved in general, in the sense that no formulas exist for obtaining its solution in all cases.

We are interested mainly in finding solutions of the differential equations of certain types. Our main purpose is to acquire technical facility, we may completely disregard certain properties like continuity, differentiability etc.

The simplest first-order ODE can be $\frac{dy}{dx} = 0$, which indicates that we are looking for such a y which has a vanishing slope. This happens for $y = \text{constant}$, i.e., a family of straight lines parallel to the x -axis.



Similarly a constant slope condition is $\frac{dy}{dx} = c$, from which we can obtain y by

$$y = c \int dx + d = y = cx + d,$$

which is a straight line with a slope c and y -intercept d .

As a next step, consider

$$\frac{dy}{dx} = f(x), \tag{6}$$

which can be solved by writing

$$y = \int f(x)dx + c.$$

However, in general

it may not be that easy to evaluate $\int f(x)dx$ since $f(x)$ may not be in an easy form.



The simplest of first-order equations is that in which the variables x and y are separable:

$$\frac{dy}{dx} = f(x, y) \equiv g(x)h(y). \quad (7)$$

For equation (7),

we only have to write this in separated form $dy/h(y) = g(x)dx$ and integrate:

$$\int \frac{dy}{h(y)} = \int g(x)dx + c.$$



Example:

Solve by separation of variables: $\frac{dy}{dx} = \frac{y}{x-1}$.

Solution:

Separating the variables,

$$\begin{aligned}\int \frac{dy}{y} &= \int \frac{dx}{x-1} + c \\ \Rightarrow \ln y &= \ln(x-1) + \ln C \\ \Rightarrow y &= C(x-1),\end{aligned}$$

which gives a family of straight lines since C is arbitrary.

Introducing a condition $y(2) = 1$, i.e.,

if the line passes through the point $(2,1)$, then we get $C = 1$ and hence the solution satisfying $y(2) = 1$ is $y = x - 1$.

It is clear that different conditions $y(a) = b$ will give us different straight lines.



At the next level of complexity is the homogeneous equation:

A function $f(x, y)$ is called **homogeneous of degree n** if

$$f(tx, ty) = t^n f(x, y)$$

for all suitably restricted x, y and t .

Thus $x^2 + y^2$, $\sqrt{x^2 + y^2}$ and $\sin(x/y)$ are homogeneous of degrees 2, 1 and 0, respectively.

The differential equation

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be **homogeneous** if M and N are homogeneous functions of the same degree.

This equation then can be written in the form

$$\frac{dy}{dx} = -\frac{M(x, y)}{N(x, y)} \equiv f(x, y)$$

Solution Procedure:

Let $y = v(x)x$ so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$. This means the substitution of the dependent variable y by v .

Then, the given equation becomes a differential equation in v . Solving for v and then replacing $v = y/x$ gives the solution for y .

Example:

Verify that the following equation is homogeneous and solve it:

$$(x^2 - 2y^2) dx + xy dy = 0.$$

Solution: The differential equation can be written as

$$\frac{dy}{dx} = 2\frac{y}{x} - \frac{x}{y}.$$

Using $y = vx$:

$$v + x \frac{dv}{dx} = 2v - \frac{1}{v}$$

$$\Rightarrow x \frac{dv}{dx} = v - \frac{1}{v}$$

$$\Rightarrow \frac{v dv}{v^2 - 1} = \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \ln(v^2 - 1) = \ln x + \ln c$$

$$\Rightarrow \sqrt{v^2 - 1} = cx.$$

Putting back $v = y/x$, the solution is obtained as

$$\sqrt{y^2 - x^2} = cx^2.$$



Equations reducible to homogeneous form

Even if a first-order differential equation is not a homogeneous one, its some specific form can be converted into a homogeneous equation. We will illustrate it by examples

Consider the following first-order equation:

$$\frac{dy}{dx} = \frac{y - x + 1}{y + x + 5}, \quad (8)$$

which is obviously not a homogeneous equation.

Step 1:

We know that $y - x + 1 = 0$ and $y + x + 5 = 0$ are two straight lines intersecting at $x = -2, y = -3$. Now, make the substitutions:

$$x = X - 2, \quad y = Y - 3. \quad (9)$$

This amounts to taking new axes parallel to the old one with $(-2, -3)$ as the new origin. It leads to

$$y - x + 1 = Y - X, \quad y + x + 5 = Y + X.$$

At the same time, $dx = dX$ and $dy = dY$. Hence, equation (8) becomes

$$\frac{dY}{dX} = \frac{Y - X}{Y + X}, \quad (10)$$

which is now a homogeneous equation.

Taking $Y = VX$, we get

$$X \frac{dV}{dX} = -\frac{1 + V^2}{V + 1}.$$

On solving

$$2 \tan^{-1} V + \ln(1 + V^2) + 2 \ln X = 2C.$$

Since $Y = VX$, now we get

$$2 \tan^{-1} \left(\frac{Y}{X} \right) + \ln(X^2 + Y^2) = 2c.$$

Again, since $X = x + 2$, $Y = y + 3$, we ultimately get

$$2 \tan^{-1} \left(\frac{y+3}{x+2} \right) + \ln\{(x+2)^2 + (y+3)^2\} = 2c,$$

which is the required solution of equation (8).

Example: Now we consider a different form of non-homogeneous equation

$$\frac{dy}{dx} = \frac{y - x + 1}{y - x + 5}. \quad (11)$$

In equation (11), we observe a common quantity $y - x$ in both numerator and denominator of the right side.

Take $y - x = u$. It renders equation (11) to

$$\frac{du}{dx} + 1 = \frac{u+1}{u+5}, \quad (12)$$

ultimately giving

$$\frac{du}{dx} = -\frac{4}{u+5}. \quad (13)$$

Equation (13) is in variables separation form

$$(u + 5)du + 4dx = 0.$$

Its solution can be obtained as

$$u^2 + 10u + 8x = 2c.$$

But since $u = y - x$, the final form of the solution is obtained as

$$(y - x)^2 + 10(y - x) + 8x = 2c.$$



Exact equations

If we begin with a family of curves $f(x, y) = c$, then its differential equation can be written in the form $df = 0$ or

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0.$$

For example, the family $x^2y^3 = c$ has $2xy^3 dx + 3x^2y^2 dy = 0$ as its differential equation.

Now let's turn around the situation and begin with the differential equation

$$M(x, y) dx + N(x, y) dy = 0. \quad (14)$$

If there happens to be a function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = M \quad \text{and} \quad \frac{\partial f}{\partial y} = N, \quad (15)$$

then (14) can be written in the form

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0 \quad \text{or} \quad df = 0.$$

Its general solution is

$$f(x, y) = c.$$

In this case, the expression $M dx + N(x, y) dy$ is said to be an **exact differential**, and (14) is called an **exact differential equation**.

For simple differential equations, it is possible to determine the exactness and find the function by inspection. Consider the following two differential equations:

$$y dx + x dy = 0, \quad \frac{1}{y} dx - \frac{x}{y^2} dy = 0.$$

The left sides of these equations can easily be recognized as the differentials of xy and x/y , respectively. Hence, the general solutions of these equations can be written as $xy = c$ and $x/y = c$.

But, in general, the exactness of all equations cannot be ascertained by a mere inspection, and we have to do some tests of exactness and also to find a method for finding the function f .



Suppose equation (14) is exact so that there exists a function f satisfying equations (15). We know that the mixed second-order partial derivatives of f can be considered to be equal:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}. \quad (16)$$

This yields

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad (17)$$

So, equation (17) is a necessary condition for the exactness of (14). Let's take an example to see how we solve an exact equation.

Example:

$$e^y dx + (xe^y + 2y)dy = 0.$$

Solution:

Here, we have $M = e^y$ and $N = xe^y + 2y$. So, $\frac{\partial M}{\partial y} = e^y$, and $\frac{\partial N}{\partial x} = e^y$.



Hence, since the condition of exactness is satisfied, the equation is exact and there exists a function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = e^y \quad \text{and} \quad \frac{\partial f}{\partial y} = xe^y + 2y.$$

Integrating the first equation with respect to x gives

$$f = \int e^y dx + g(y) = xe^y + g(y),$$

so that

$$\frac{\partial f}{\partial y} = xe^y + g'(y).$$

Since this partial derivative must also equal $xe^y + 2y$, we have $g'(y) = 2y$. Therefore, $g(y) = y^2$ giving us $f = xe^y + y^2$.

Hence,

$$xe^y + y^2 = c$$

is the desired solution to the given differential equation.